

1 AISAnalyze: streamlining AIS vessel data for 2 environmental analyses in R

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Software

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10 Summary

11 AISAnalyze is an open-source R package that provides an end-to-end workflow for transforming
12 Automatic Identification System (AIS) vessel tracking data into datasets tailored for
13 environmental analyses. AIS is an international vessel tracking system that continuously
14 broadcasts ship identity, position, speed, heading, and other navigational information for
15 maritime safety. Beyond its original purpose, AIS data have become an essential resource
16 for studying maritime traffic in disciplines including marine ecology, conservation, fisheries,
17 transportation, and marine spatial planning ([Crum et al., 2019](#); [Robards et al., 2016](#); [Yang et al., 2019](#)).

18 Despite their widespread use, AIS datasets frequently contain millions of observations together
19 with transmission delays, positional inaccuracies, duplicated records, and incomplete vessel
20 information ([Nachtsheim et al., 2023](#); [Natale et al., 2015](#); [Robards et al., 2016](#)). Preparing
21 these data for scientific analyses is therefore computationally demanding and often requires
22 substantial preprocessing before vessel trajectories can be reliably analysed ([Kiersztyn et al., 2025](#)).

23 AISAnalyze provides an integrated workflow that implements the principal preprocessing
24 operations required for scientific analyses of vessel movements, including travel estimation,
25 correction of GPS errors and transmission delays, identification of non-vessel signals (e.g. base
26 stations and aircraft), trajectory interpolation, extraction of vessel positions around user-defined
27 locations and times, and summarization of vessel characteristics. The package is organized
28 around a small number of user-facing functions that preserve a consistent data structure
29 throughout the workflow, facilitating integration into existing R analysis pipelines.

30 AISAnalyze is intended for researchers and practitioners working with large AIS datasets,
31 particularly in marine ecology, environmental impact assessment, underwater noise studies,
32 and marine spatial planning. As the use of AIS data continues to expand across scientific
33 and management applications ([Yang et al., 2019](#)), the package provides a reproducible and
34 computationally efficient framework for preparing AIS data for downstream statistical and
35 spatial analyses. AISAnalyze is openly available on GitHub, archived on Zenodo, and distributed
36 under the MIT license.

39 Statement of need

40 Maritime traffic is among the most pervasive human activities in the world oceans and
41 contributes to a wide range of environmental pressures, including underwater noise, collision

42 risk with marine wildlife, habitat disturbance, and greenhouse gas emissions (Dekeling et
43 al., 2014; Frisk, 2012; Halpern et al., 2008, 2015). Quantifying the spatial and temporal
44 distribution of vessel traffic has therefore become an essential component of marine ecological
45 research, environmental impact assessments, and marine spatial planning (Benhemma-Le Gall
46 et al., 2023; Pigeault et al., 2024; Pirotta et al., 2025; Plot et al., 2025; Robards et al., 2016).
47 AIS data provide detailed information on vessel movements over broad spatial and temporal
48 scales and are increasingly combined with environmental observations, species distributions,
49 animal telemetry, and oceanographic data to investigate the effects of maritime traffic on
50 marine ecosystems and to support evidence-based management decisions (Yang et al., 2019).

51 Many of these applications require vessel trajectories to be matched with external datasets
52 (e.g. animal trajectory), creating preprocessing requirements that extend beyond trajectory
53 reconstruction alone. However, raw AIS data are not directly suitable for scientific analyses:
54 transmission delays, positional inaccuracies, duplicated records, missing vessel information, and
55 non-vessel signals require substantial preprocessing before vessel trajectories can be reliably
56 analysed (Kiersztyn et al., 2025; Robards et al., 2016). Modern AIS datasets frequently contain
57 tens to hundreds of millions of observations (Natale et al., 2015), making routine preprocessing
58 computationally demanding and often requiring researchers to develop workflows that are
59 difficult to reproduce and maintain.

60 To date, these preprocessing tasks were commonly implemented through project-specific
61 scripts that were repeatedly redeveloped across independent studies (Benhemma-Le Gall
62 et al., 2023; Nachtsheim et al., 2023; Paillé et al., 2024). Project-specific scripts reduce
63 reproducibility, increase development time, and create unnecessary barriers for researchers
64 wishing to integrate maritime traffic into ecological or environmental analyses. A standardized
65 preprocessing framework therefore benefits both software reuse and the reproducibility of
66 scientific studies relying on AIS data. AISanalyze was developed to simplify this preprocessing
67 stage by providing an integrated workflow that transforms raw AIS data into analysis-ready
68 datasets while preserving compatibility with further statistical and spatial analyses. By reducing
69 the technical barriers associated with large AIS datasets, the package facilitates reproducible
70 studies of maritime traffic for researchers, environmental practitioners, and policymakers working
71 across disciplines where vessel movements constitute an important explanatory variable.

72 State of the field

73 The increasing use of AIS data across scientific disciplines has stimulated the development
74 of software tools dedicated to vessel trajectory processing, movement reconstruction, and
75 maritime traffic analysis (Robards et al., 2016). Until recently, AIS preprocessing was largely
76 performed through project-specific scripts, limiting reproducibility and requiring researchers to
77 repeatedly implement similar correction and interpolation procedures. Over the past few years,
78 several open-source tools have been developed to facilitate AIS analyses by addressing specific
79 aspects of vessel trajectory processing, including database management, anomaly detection,
80 trajectory reconstruction, and movement modelling (Magnussen et al., 2023; Nabe-Nielsen et
81 al., 2026; Park et al., 2026; Spadon et al., 2024; Takahashi et al., 2024). These tools provide
82 valuable capabilities for their intended applications but often require users to combine multiple
83 software packages or develop additional scripts to complete a typical scientific workflow (Nabe-
84 Nielsen et al., 2026; Spadon et al., 2024). Furthermore, some reconstruction methods rely on
85 historical vessel behaviour or predictive models to estimate vessel positions (Magnussen et al.,
86 2023; Park et al., 2026; Takahashi et al., 2024). While these approaches are appropriate for
87 many applications, they introduce modelling assumptions that may not be desirable when the
88 objective is to preserve the original vessel observations while preparing datasets for subsequent
89 ecological or environmental analyses.

90 AISanalyze was developed to address a complementary need. Rather than focusing on the
91 development of a new trajectory reconstruction algorithm, the package provides a reproducible
92 preprocessing workflow that prepares AIS datasets for further scientific analyses. It integrates the

principal preprocessing operations commonly required by marine ecologists and environmental scientists within a single software package, including travel estimation, GPS correction, identification of non-vessel signals, trajectory interpolation, extraction of vessel positions around user-defined locations and times, and retrieval of vessel characteristics. This allows researchers and practitioners to integrate AIS data with their own environmental datasets within a single, computationally efficient framework. A key design choice of AISanalyze is the use of deterministic preprocessing algorithms that can be readily interpreted by users. For example, vessel trajectories are interpolated using linear interpolation between consecutive AIS positions, following approaches adopted in several existing AIS processing frameworks (Nabe-Nielsen et al., 2026; Park et al., 2026; Spadon et al., 2024). This choice minimizes methodological assumptions, preserves transparency, and remains computationally efficient for large AIS datasets. AISanalyze therefore complements existing AIS software by providing an integrated workflow that reduces the need for project-specific preprocessing scripts while facilitating reproducible ecological and environmental analyses.

Software design

AISanalyze was intentionally designed around a small number of user-facing functions representing the principal stages of AIS preprocessing. This modular workflow reduces the learning curve while allowing users to combine only the processing steps required for a given study.

A primary design objective was to minimise computational time when processing large AIS datasets. Computationally intensive functions therefore implement parallel processing where appropriate and rely on vectorised algorithms operating directly on tabular data, avoiding repeated manipulation of spatial objects whenever possible. Consequently, large AIS datasets can typically be transformed from raw vessel positions to analysis-ready data in only a few seconds to minutes. These design choices substantially reduce execution time and memory usage.

A second design objective was to preserve compatibility with the broader R ecosystem and established spatial analysis workflows. All functions return standard data frames that retain the original data structure while appending AIS-derived variables, facilitating downstream statistical, spatial, and ecological analyses without requiring users to modify existing workflows.

Transparency and reproducibility were also central design considerations. AISanalyze implements deterministic preprocessing algorithms whose behaviour can be directly interpreted and reproduced. For example, vessel trajectories are reconstructed using linear interpolation, avoiding additional modelling assumptions while preserving observed vessel movements. Similarly, GPS correction procedures rely on explicit movement constraints that remain fully reproducible across analyses.

Research impact statement

AISanalyze supports past and ongoing peer-reviewed studies (Maurer et al., 2026; Pigeault et al., 2024; Wynn-Simmonds et al., 2025) and collaborative research projects (*HabitatWal*, 2020-2026; *SATURN*, 2020-2024; *SustainMare*, 2021-2026) investigating the ecological impacts of maritime traffic on marine ecosystems.

The package has been used to investigate the short-term relationship between maritime traffic and harbour porpoise distribution throughout the North Sea between 2015 and 2022 by combining extensive AIS records with aerial survey data (Pigeault et al., 2024). It has also been employed to quantify the spatial overlap between harbour seals tracked by telemetry and maritime traffic in the English Channel, enabling the integration of vessel trajectories with habitat selection analysis (Wynn-Simmonds et al., 2025). More recently, AISanalyze was used

to investigate the response of harbour seals to their local environment, including vessel presence and activity, in the German Wadden Sea (Maurer et al., 2026). These studies illustrate the package ability to efficiently preprocess large AIS datasets and integrate vessel trajectories with biological observations, environmental data, and spatial analyses across broad spatial and temporal scales.

Beyond these applications, AISanalyze is currently used in collaborative research initiatives, including *HabitatWal* (2020–2026) and *SustainMare* (2021–2026). These projects investigate the effects of maritime activities on marine biodiversity, underwater radiated noise, and ecosystem management, and use AIS data to support evidence-based conservation and marine spatial planning. The package has also supported research conducted within the *SATURN* (2020–2024) project and related ongoing studies, which develops scientific knowledge and practical solutions to understand and reduce underwater radiated noise from shipping and its impacts on aquatic species.

By providing a transparent, reproducible, and computationally efficient preprocessing workflow, AISanalyze reduces the technical barriers associated with analysing large AIS datasets and facilitates the integration of maritime traffic data into ecological and environmental research. This enables reproducible analyses of human pressures on marine ecosystems and helps inform evidence-based marine spatial planning and conservation management.

AI usage disclosure

Generative artificial intelligence (AI) tools were used to improve the grammar of the manuscript.

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